

MODEL QUESTION PAPER

TED (10) – 4056

(Revision – 2010)

FIFTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY

DIGITAL AND DATA COMMUNICATION SYSTEMS

(EC)

(Time : 3 hours)

(Maximum marks:100)

PART – A

(Answer *all* questions in one or two sentences. Each question carries 2 marks)

I

1. Define the term information rate
 2. List two types of pulse time modulation schemes
 3. Write the equation for maximum dynamic range in dB of a PCM system
 4. Calculate the number of levels available in PCM if number of bits per sample is 8
 5. List the two architectures commonly used for ISDN
- 2x5=10

PART – B

(Answer any *five* of the following questions. Each question carries 6 marks)

II

1. Describe CRC method of error correction
 2. Compare PAM with PWM
 3. Distinguish between Delta modulation and Adaptive Delta modulation
 4. Discuss Phased and slotted operation for time division time switching
 5. Describe load sharing mod operation for centralised SPC
 6. Draw the circuit for generation of PPM using IC 555 and Explain
 7. Describe Shannon-Hartly theorem
- 5x6=30

PART – C

(Answer *one* full question from each module. Each question carries 15 marks)

MODULE – I

- I 1. Derive the equation for finding entropy 7
2. Telephone line has a bandwidth of 4.2Khz and signal to noise ratio of 40dB. A signal is transmitted using 8 level code calculate the maximum theoretical data rate 8

OR

- II 1. Explain block interleaving and convolution interleaving methods for burst error channel 10
2. Describe Hamming code for error correction 5

MODULE – II

- III 1. Compare synchronous and asynchronous methods of data transmission 6
2. Draw the block diagram of FDM and explain each block 9

OR

- IV 1. Explain generation of PWM using suitable diagram 8
2. State the sampling theorem. write its significant in data communication 7

MODULE – III

- V 1. With block diagram Explain PCM Transmitter and receiver 9
2. Draw the block diagram adaptive delta modulation transmitter and explain the working 6

OR

- VI 1. Illustrate the need for companding 5
2. Mention the limitations of PCM system. Discuss how it is overcome in Differential PCM 10

MODULE – IV

- VII 1. Draw and explain the working of two motion selector switch used in strowger exchange 10
2. Compare centralised SPC and Distributed SPC 5

OR

- VIII 1. Draw and explain block diagram of time division time switching 10
2. List out advantages of electronic exchange compared to crossbar exchange 5

